CLONING ENCRYPTED QUBITS

You Can’t Clone a Qubit – Unless You Encrypt It First

Quantum Physics · Information Theory · Cryptography

CRYPTORAVE BR 2026 · São Paulo, Brasil · Jullyano Lino · github.com/jullyanolino/encrypted_qubit_cloning

$ whoami

name

Jullyano Lino

role

(Quantum) Cybersecurity Researcher/Consultant

affiliation

IADB | UNINTER | SENAI/CIMATEC

bio

Computer Science, Math, Quantum Communication.
Quantum cybersecurity evangelist, speaker, developer & researcher.

He replicated the encrypted quantum cloning protocol (Yamaguchi & Kempf) on IBM Heron R2 hardware (n=2 to n=15) & identified 4 attack scenarios: quantum ransomware, HNDL, quantum steganography, and security reduction.

GitHub Repo

Theory Paper

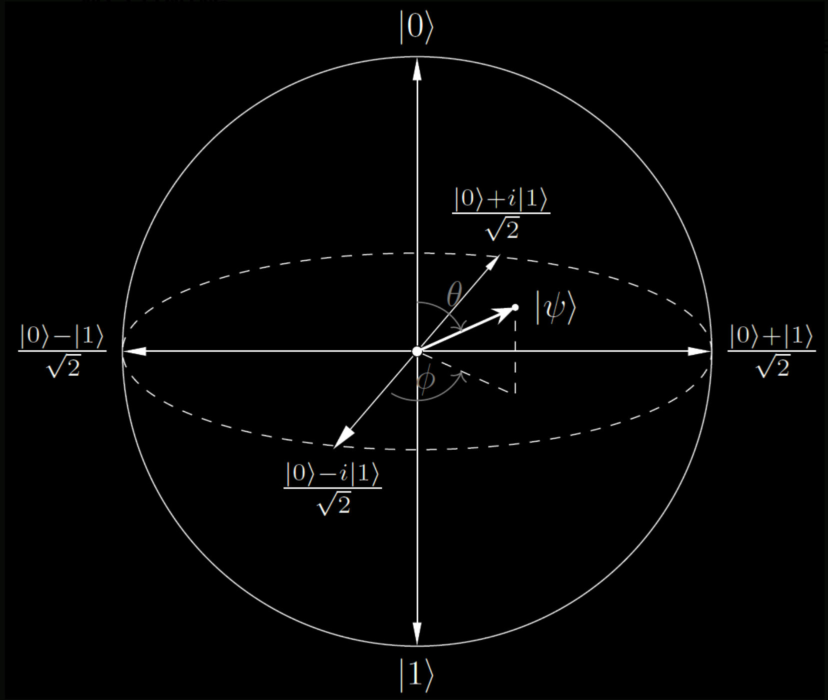
Demo Paper

Social Hub

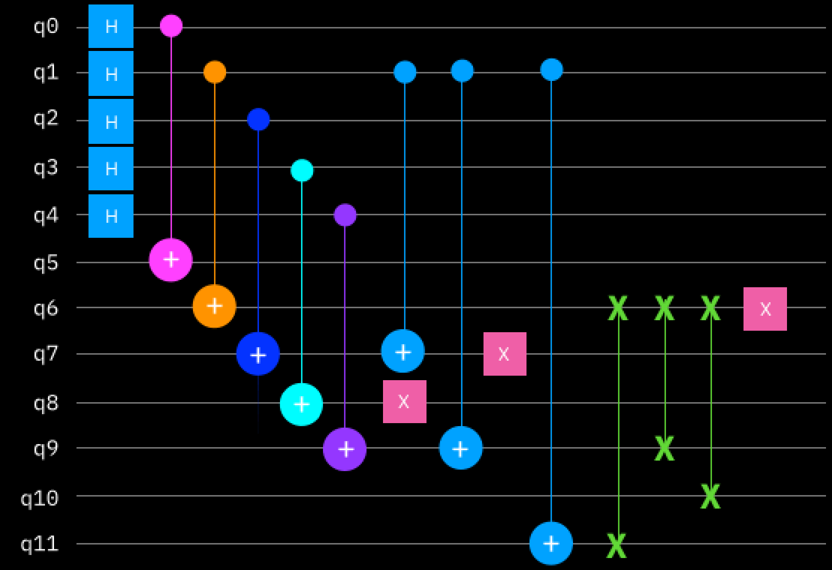
d7tad7audops7397o1dg

github.com/jullyanolino/encrypted_qubit_cloning · arxiv.org/abs/2501.02757 · arxiv.org/abs/2602.10695
quantum.ibm.com/jobs/d7tad7audops7397o1dg

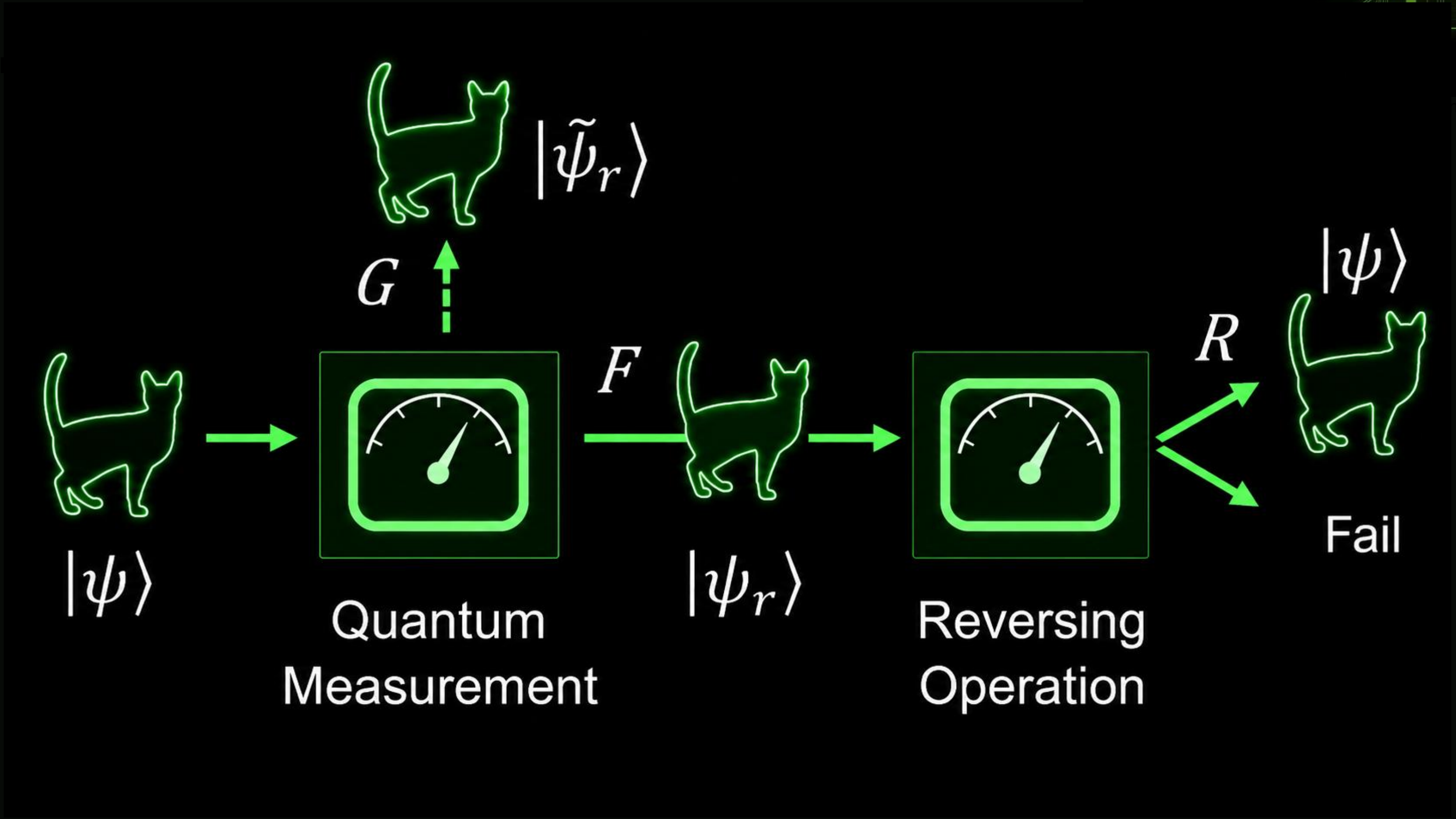
[deep dive] Quantum Computing 101



Compute the depth of the circuit by playing Tetris



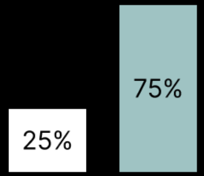
[deep dive] Quantum Computing 101



QUANTUM INTERFERENCE

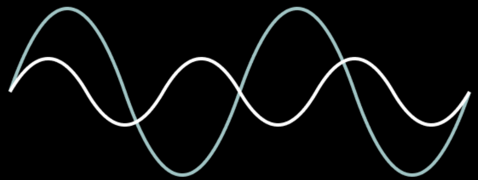
CLASSICAL

Classical computation **can only represent probabilities that sum to one**



QUANTUM

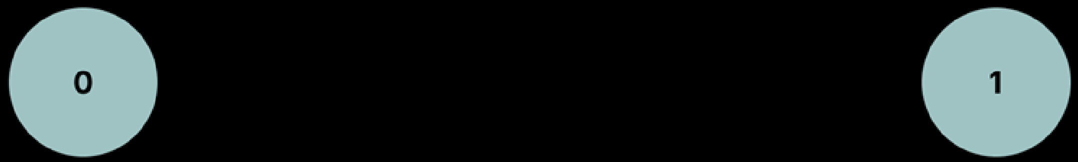
Quantum states are waveforms that have **positive** and **negative** amplitudes that can **combine to amplify the right answer**



QUANTUM SUPERPOSITION

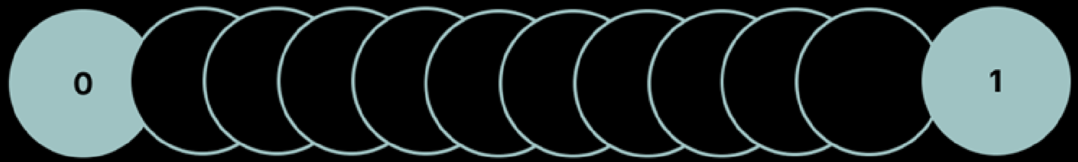
CLASSICAL

Classical bits may take the value of **either** one **or** zero



QUANTUM

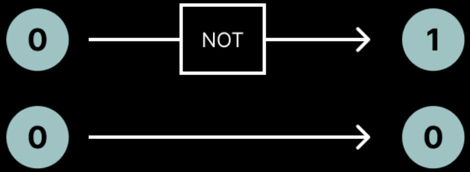
Quantum superposition enables qubits to represent **many states at once**



QUANTUM ENTANGLEMENT

CLASSICAL

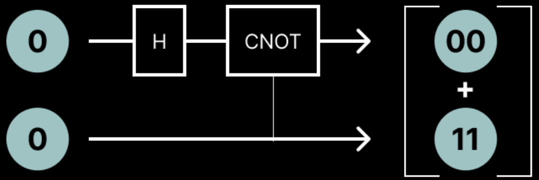
Classical bits **are logically, not physically, dependent** on other bits



These two bits are independent

QUANTUM

The **state of one qubit** in an entangled pair **can affect the other, even separated by distance**



Measuring the state of one bit defines the other, called a Bell state

[deep dive] Quantum Mechanics 101

No-Cloning, No-Signaling, Holevo, Monogamy

NO-CLONING

1982 · Wootters & Zurek

$\nexists\, U:\ U|\psi\rangle_A|0\rangle_B = |\psi\rangle_A|\psi\rangle_B$ for all $|\psi\rangle$

INTUITION

Cannot copy unknown quantum state.

ENC CLONING:

RESPECTED ✓
Clones $\{S_i\}$ are $I/2$, not $|\psi\rangle$.
Only one recoverable via $\{N_i\}$.

NO-DELETING

2000 · Pati & Braunstein

$\nexists\, U:\ U|\psi\rangle_A|\psi\rangle_B = |\psi\rangle_A|0\rangle_B$ for all $|\psi\rangle$

INTUITION

Cannot erase one copy if you have two.

ENC CLONING:

RESPECTED ✓
U_dec consumes $\{N_i\}$, not the signal $\{S_i\}$.

NO-SIGNALING

1981 · Ghirardi et al.

$P(a|x, \lambda)$ independent of choice y at distant lab B.

INTUITION

Entanglement cannot transmit information.

ENC CLONING:

RESPECTED ✓
$\chi(\{Si\})=0$ bits.
No info in transit.

HOLEVO BOUND

1973 · Holevo

$\chi(\rho_X;\ M) \leq S(\rho)-\Sigma p_i S(\rho_i) \leq \log d$

INTUITION

Max classical bits extractable from qubits.

ENC CLONING:

SATURATED ✓
$\chi=S(I/2)-S(I/2)=0$.
Absolute floor hit.

MONOGAMY of ENTANGLEMENT

2000+ · Coffman et al.

$E(A:B) + E(A:C) \leq E(A:BC)$

INTUITION

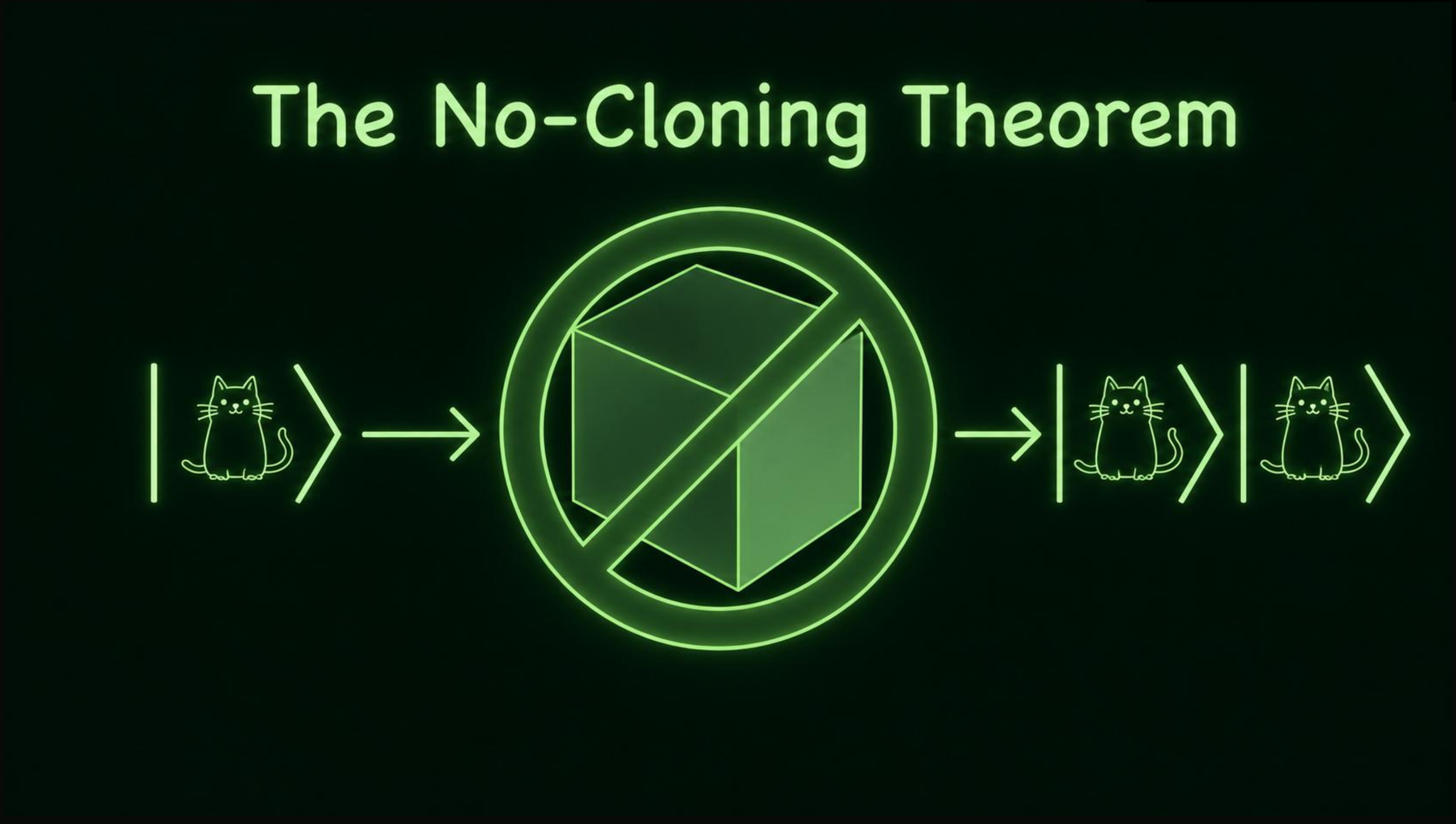
Entanglement cannot be freely shared.

ENC CLONING:

LEVERAGED ✓
$\{N_i\}$ entangled with all $\{S_i\}$ simultaneously.

Encrypted cloning does not violate any of these theorems — it operates at their exact intersection, exploiting entanglement structure within the allowed bounds.

You CAN'T clone qubits.
Maybe cats.



ACT 1 THE PROTOCOL

Physics first, intuition always

01 COLD OPEN
30s

Terminal output first.
No slides yet.
Paradox delivered.

02 THE RECEIPTS
45s

PRL + arXiv + GitHub.
Verifiable job IDs.
Every number grounded.

03 THE IMPOSSIBLE
90s

3 properties. Can't
coexist. They do.
Venn diagram reveal.

04 THE HEIST
3 min

INIT → ENCRYPT
→ DECRYPT.
Noise qubits NEVER
TOUCHED ← key line.

05 PERFECT STEALTH
90s

Why $\rho = I/2$.
Bloch sphere: avg of
4 Bell states = center.
Holevo: 0 bits.

06 THE THRESHOLD
90s

Step function graph.
$k < n \rightarrow Fe = 0.25$.
$k = n \rightarrow Fe = 1.00$.
3s silence at jump.

KEY CLAIM: Encrypted cloning is unitary · $\rho_{Si} = I/2$ · Holevo = 0 bits ·
All-or-nothing key (n qubits) · Verified IBM Heron R2

```
$ python enc_qubit_cloning.py --verify
```

Fidelity: 1.000 ± 0.000 ✓ Perfect cloning verified

Perfect Cloning:	**FORBIDDEN**	✗
Encryption:	**REQUIRED**	✓
Detection:	**IMPOSSIBLE**	✓

"Three things that cannot coexist. They do."

THE CODE

enc build_uenc_matrix(n)

```
def build_uenc_matrix(n: int) -> np.ndarray:
    X = np.array([[0,1],[1,0]], dtype=complex)
    Z = np.array([[1,0],[0,-1]], dtype=complex)
    dim = 2 ** (n + 1)
    Xall = X.copy()
    Zall = Z.copy()
    for _ in range(n):
        Xall = np.kron(Xall, X)   # σ₁^⊗(n+1)
        Zall = np.kron(Zall, Z)   # σ₃^⊗(n+1)
    c, s = np.cos(np.pi/4), np.sin(np.pi/4)
    UX = c*np.eye(dim) - 1j*s*Xall   # exp(-iπ/4 σ₁^⊗(n+1))
    UZ = c*np.eye(dim) - 1j*s*Zall   # exp(-iπ/4 σ₃^⊗(n+1))
    return UX @ UZ
```

GATE COUNT (linear in n)

Encrypt: 4n CZ, depth 2n

Decrypt: 15n+7 CZ, depth ~5n

dec build_udec_matrix(n)

```
def build_udec_matrix(n: int) -> np.ndarray:
    PAU   = [I2, X, Y, Z]
    PAU_T = [I2, X, -Y, Z]
    alpha = {0:1, 1:1j,
             2:-(1j)**(n+1), 3:1j}
    phi_plus = [1,0,0,1]/√2
    U = zeros((2^(n+1), 2^(n+1)))
    for μ in range(4):
        ϕ_μ  = (σ_μ ⊗ I)|ϕ+⟩
        proj = |ϕ_μ⟩⟨ϕ_μ|
        tail = PAU_T[μ]^⊗(n-1)
        U += α_μ · (proj ⊗ tail)
    return U
```

STRUCTURE

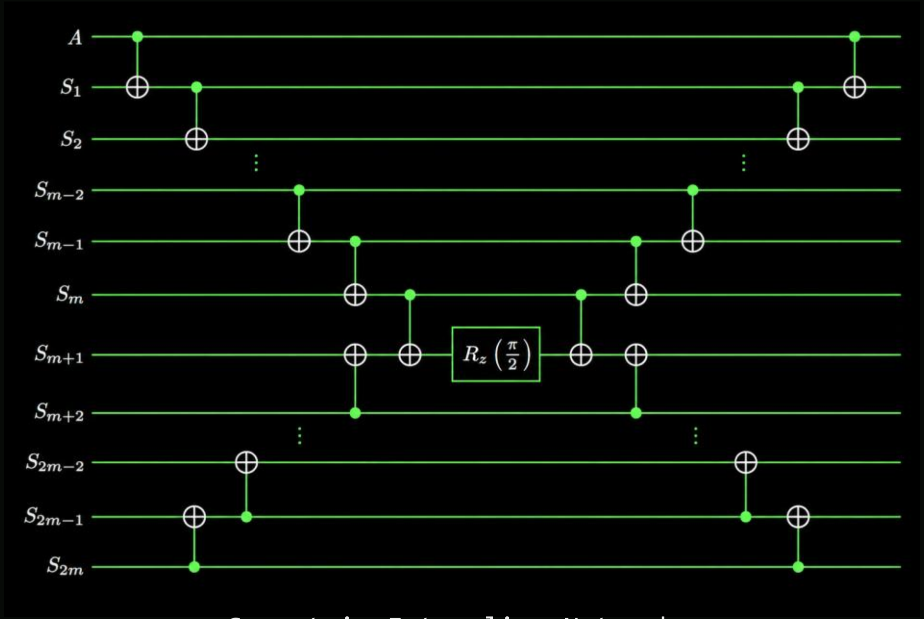
IBM Heron R2: {rz, sx, x, cz} | transpiler: Qiskit opt-3

Total: ~21n+11 two-qubit gates

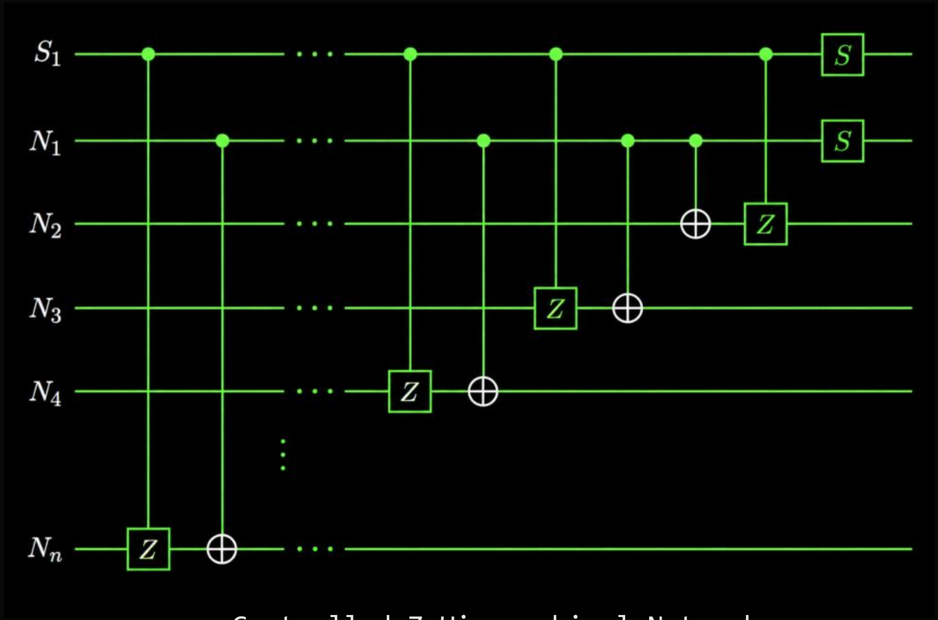
THE CIRCUITS

enc `build_uenc_matrix(n)`

dec `build_udec_matrix(n)`

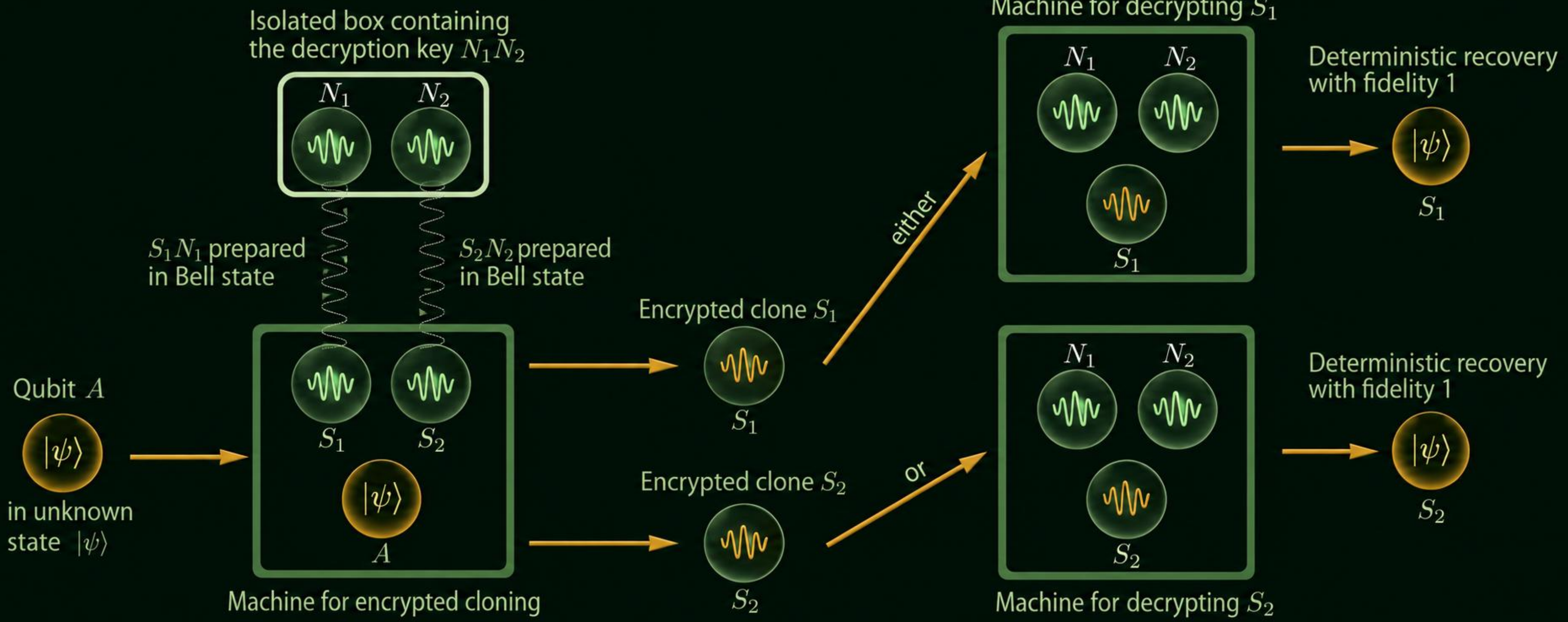


Symmetric Entangling Network
→
Controlled phase propagation



Controlled-Z Hierarchical Network
→
Scalable distributed entanglement

Cloning Encrypted Qubits!



THE IMPOSSIBLE

Three properties that cannot coexist — until now.

PERFECT COPIES

$Fe = 1.000$ when decrypted
All n clones recover $|\psi\rangle$ exactly

PURE NOISE

$\rho_{Si} = I/2$ individually
Holevo bound = 0 extractable bits

UNITARY PROCESS

No measurements taken
Fully reversible, no collapse

→ INTERSECTION ←

Classical: copies are perfect but observable.

UQCM: copies are imperfect ($Fe \leq 5/6$) but unobservable.

Encrypted cloning: copies are perfect AND unobservable — until you decrypt.

THE HEIST

Setup → Execute → Escape. Three beats.

01 INIT

$|\psi\rangle_A \quad \otimes \quad |\phi\rangle^{\otimes n}$

n Bell pairs
$|\phi\rangle_{\{S,N\}} = (|00\rangle+|11\rangle)/\sqrt{2}$

n quantum one-time pads.

$\rho_{Si} = I/2$ locally

Perfectly correlated globally with $\{N_i\}$.

02 ENCRYPT

U_{enc} acts on $A \otimes \{S_i\}$

Noise qubits $\{N_i\}$:
NEVER TOUCHED ←

After U_enc:
$\rho_{Si} = I/2 \quad \forall i$
$\rho_A = I/2$

Zero extractable bits (Holevo bound)

03 DECRYPT

U_{dec} on $S_j \otimes \{N_i\}$
(ALL n noise qubits)

Result:
Fe = 1.000 (ideal)
Fe = 0.762 (hardware)

Cost: $\{N_i\}$ consumed.

Remaining S_i remain $I/2$. Single-use key.

Circuit cost: Encrypt → 4n CNOT, depth 2n | Decrypt → 15n+7 CNOT, depth ~5n | Total: ~21n CNOT (linear scaling)

FORMAL SPECIFICATION

Complete protocol definition.

INPUT

Qubit A in unknown state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$
n Bell pairs: $|\phi\rangle_{\{S_i N_i\}} = (|00\rangle + |11\rangle) / \sqrt{2}$

ENCRYPTION

U_enc on A and $\{S_i\}$ · $\{N_i\}$ untouched

$$U_{enc} = \exp(-\pi i/4 \cdot \sigma_1^{(A)} \otimes \bigotimes_i \sigma_1^{(S_i)}) \cdot \exp(-\pi i/4 \cdot \sigma_3^{(A)} \otimes \bigotimes_i \sigma_3^{(S_i)})$$

Result:
- $\rho_{\{S_i\}} = I/2$ for all i (state-independent)
- $\rho_A = I/2$ (maximally mixed after encoding)
- $\{N_i\}$: unchanged (never touched A)

DECRYPTION

U_dec on chosen S_j and all $\{N_i\}$

Recovers $|\psi\rangle$ on S_j with Fe = 1, exactly.
Cost: $\{N_i\}$ consumed. One decryption only.

GATE COUNT (linear in n)

Encryption
2q gates: 4n depth: 2n

Decryption
2q gates: 15n+7 depth: ~5n

Total
2q gates: 21n+11 depth: ~7n

HARDWARE
IBM Heron R2
Native gate set:
{rz, sx, x, id, cz}

Transpiler:
Qiskit opt-level 3

[deep dive] Fe = 0.25 | 1.00

ibm_kingston 0.239 · ibm_marrakesh 0.253 · ibm_fez 0.257 | all within ±0.02 of theory 0.250

WHAT IS Fe?

Fe = entanglement fidelity
Measures how well a quantum
channel preserves entanglement.
Fe=1 → perfect. Fe=0 → destroys.

WHAT IS k?

k = number of noise qubits
$\{N_1 \ldots N_k\}$ you possess out of n.
k is your 'partial key'.
The protocol needs ALL n.

k < n → Fe = 0.25

You're missing ≥1 noise qubit.
$\rho_{Si} = I/2$ (maximally mixed).
F_avg = 1/d = 1/2 (d=2, qubit).
Fe = ((d+1)·F_avg − 1) / d = 0.25

k = n → Fe = 1.000

You have ALL n noise qubits.
U_dec reconstructs $|\psi\rangle$ perfectly.
Fe = 1.000 (ideal) / ~0.76 (hw).
Step function — no gradual rise.

DERIVATION

$\rho_{Si} = I/2$ (after U_enc)

State fidelity to $|\psi\rangle$:
$F(\rho, |\psi\rangle) = \langle\psi|\rho|\psi\rangle$
$= \langle\psi|(I/2)|\psi\rangle$
$= 1/2$

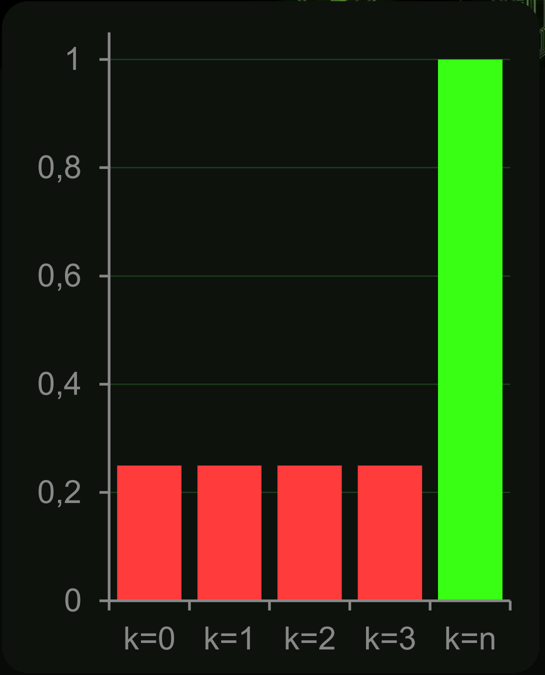
Average over all $|\psi\rangle$:
F_avg = 1/d = 1/2 (d=2)

Horodecki relation:
F_avg = (d·Fe + 1)/(d+1)

Invert for Fe:
Fe = ((d+1)·F_avg − 1) / d
= (3·(1/2) − 1) / 2
= (3/2 − 1) / 2
= (1/2) / 2

Fe = 1/4 = 0.25 ✓

This is d-dimensional minimum:
$Fe\_min = 1/d^2$ (for d=2: 0.25)



Step function — no gradual rise

0.25 is absolute minimum
for any qubit channel

WHY THE UNIVERSE HAD TO BREAK

Local Realism — and why encrypted cloning is the proof

LOCALITY

Objects are only influenced by their immediate surroundings. No influence faster than light.

REALISM

Objects have definite properties independent of measurement. "The moon exists when no one looks."

~~LOCAL REALISM = Classical World~~

⟹

BELL TESTS SAY NO

Bell's Theorem (1964):

No local hidden-variable theory can match QM predictions.

CHSH: Classical $S \leq 2$ / Quantum $S = 2\sqrt{2} \approx 2.828$

This hardware result:

$S = 2.331 \pm 0.016$ (ibm_kingston, n=2)

$S > 2$ ✓ Bell inequality violated

Nobel 2022 — Aspect, Clauser & Zeilinger:

"The universe is not locally real."

⟹

REJECT REALISM

$\rho_{Si} = I/2$

Not definite until decrypted

OR REJECT LOCALITY

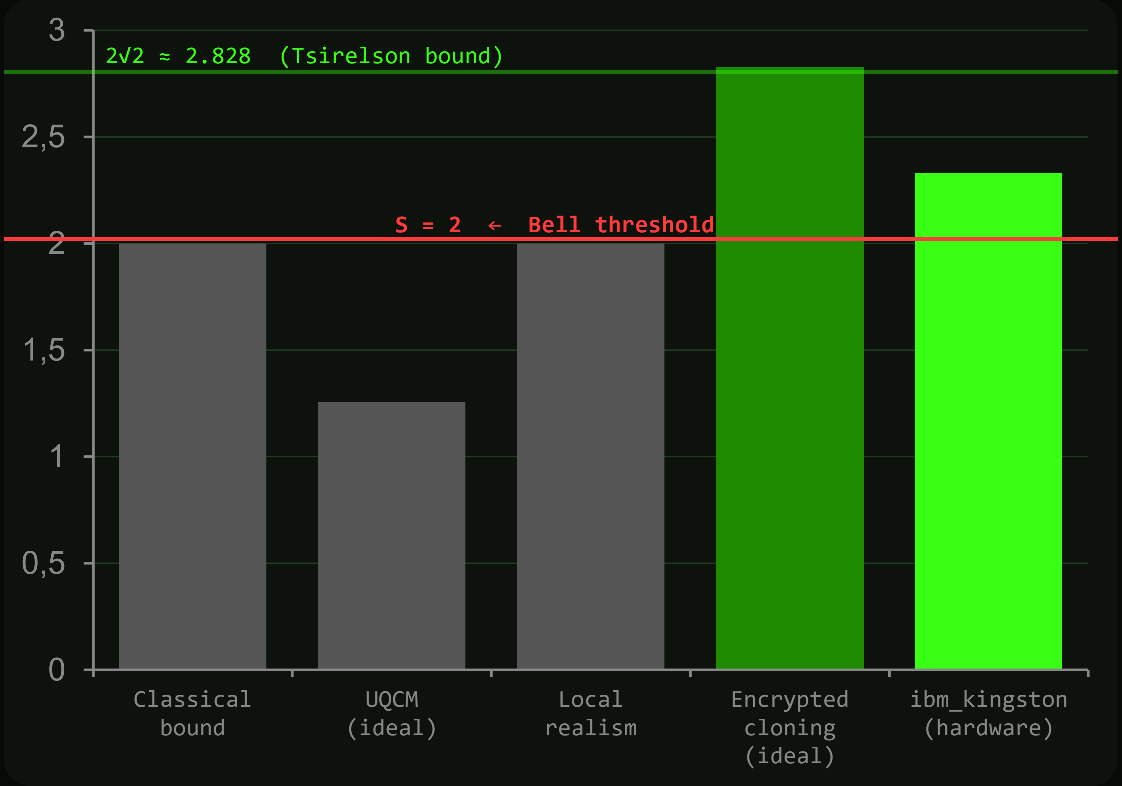
Entanglement acts at distance

Encrypted cloning exists precisely because local realism fails.

$\rho_{Si} = I/2$ is not a trick — it is the universe refusing to commit before decryption.
$F_e = 1.000$ is not a violation of no-cloning; it is the cost of forcing reality to decide.

BELL INEQUALITY VIOLATION

Decoherence destroys correlations. Encrypted cloning preserves them.



WHY THIS MATTERS

Fidelity can be inflated by shot noise or favorable calibration.

CHSH violation requires genuine non-classical correlations — decoherence cannot fake S > 2.

THE GAP

UQCM: S = 1.257 (fails ideally)
Enc. cloning: S = 2.828 (Tsirelson)
Hardware: S = 2.331 ± 0.016 ✓

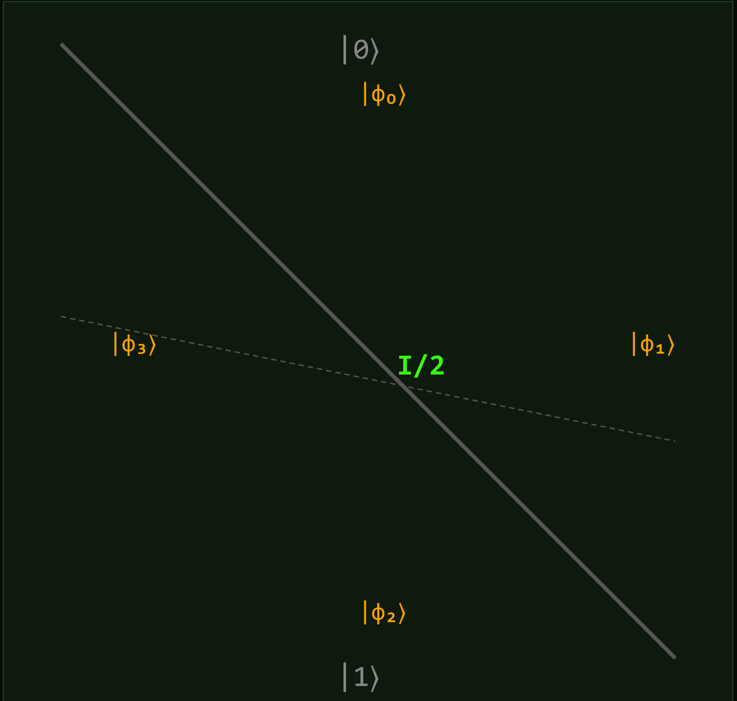
job_id: d7tad7audops7397o1dg

CONCLUSION

The entanglement is real.
The protocol behaves as predicted.
The hardware result is not an artifact.

PERFECT STEALTH

Why every encrypted clone is indistinguishable from random noise.



Average of 4 Bell states
= center of Bloch sphere
= I/2

KEY EQUATION

$\rho_{Sj} = (1/4)\ \Sigma\ |\phi_\mu\rangle\langle\phi_\mu| \ = \ I/2$

The 4 Bell states form a complete orthonormal basis.
Their equal mixture = identity.
Engineered, not accidental.

HOLEVO BOUND

$\chi(|\psi\rangle \rightarrow \text{measure } S_j) \ = \ S(I/2) - S(I/2) \ = \ 0 \text{ bits}$

Zero extractable bits.
Not 'few bits' — information-theoretically zero.
No future algorithm changes this. The information isn't hidden; it's absent.

Encrypted clone indistinguishable from:
decohered qubit | uninitialized qubit | thermal noise

No statistical test separates these cases.
quantum ransomware = undetectable!

[deep dive] $\rho_{Si} = I/2$

Indistinguishability from the Quantum Vacuum — math status after U_enc

STATE OF ρ_Si AFTER U_enc — COMPLETE MATHEMATICAL DESCRIPTION

DENSITY MATRIX

$\rho_{Si} = I/2$
$= (|0\rangle\langle 0| + |1\rangle\langle 1|)/2$

BLOCH VECTOR

$r = (\langle\sigma x\rangle, \langle\sigma y\rangle, \langle\sigma z\rangle)$
$= (0, 0, 0)$

VON NEUMANN ENTROPY

$S(\rho) = -\mathrm{Tr}(\rho \log \rho)$
$= 1$ bit (MAXIMUM)

HOLEVO CAPACITY

$\chi(\rho_{Si}; M) = 0$ bits
$\forall$ measurement M

QUANTUM VACUUM (Field Theory)

Lowest energy state $|vac\rangle$
Zero photon number $\langle n\rangle=0$
But: zero-point fluctuations
$\Delta x \cdot \Delta p = \hbar/2$

No accessible classical info
to observer without phase ref.

≠ I/2 strictly,
but same info content

THERMAL STATE (T → ∞)

ρ_thermal = e^(−βH)/Z
as β→0 (T→∞):
ρ_thermal → I/2 exactly

$\langle\sigma x\rangle=\langle\sigma y\rangle=\langle\sigma z\rangle=0$
Maximum temperature state

≡ I/2 exactly
(SAME STATE)

ENCRYPTED CLONE ρ_Si (after U_enc)

ρ_Si = I/2 by construction
(U_enc engineered this)

$\langle\sigma x\rangle=\langle\sigma y\rangle=\langle\sigma z\rangle=0$
$S(\rho)=1$ bit maximum
$\chi=0$ bits exactly

= I/2 exactly
(BY THEOREM)

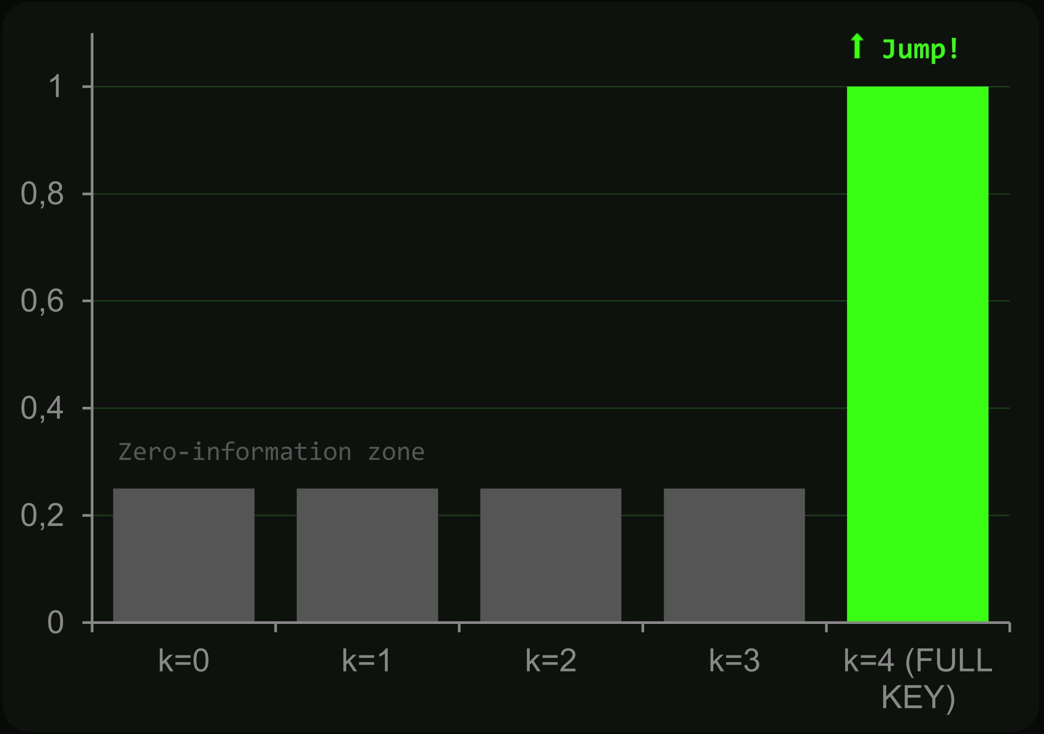
OPERATIONAL CONSEQUENCE

$D(\rho_{Si}, \rho_{thermal}(T\to\infty)) = 0$ → No measurement, no POVM, no quantum computer can distinguish an encrypted clone from thermal noise at maximum temperature. The hidden state $|\psi\rangle$ is absent — not hidden.

Indistinguishability is information-theoretic (Holevo), not computational — no future algorithm changes this.

THE THRESHOLD

Classical keys degrade. This one doesn't — until it does.



CLASSICAL (AES-256)

255/256 bits known → 2^1 guesses left
Smooth degradation

ENCRYPTED CLONING

n−1 qubits known → ∞ guesses
Information not present. Step function.

SECURITY COROLLARY

Distribute {Ni} across n parties.
Anyone refuses → data is unrecoverable.
Unanimous consent required.
No honest majority.

[deep dive] Crypto Security Model

U_enc and U_dec: what type of security, what assumptions, what gaps.

U_{enc}

✓ **Perfect Secrecy**

$H(|\psi\rangle|\{Si\}) = H(|\psi\rangle)$
(key independence — quantum Shannon analog)

✓ **IND-CPA Security**

No QPT adversary can distinguish $U_{enc}(|\psi_0\rangle)$ from $U_{enc}(|\psi_1\rangle)$ given $\{S_i\}$ only.

✓ **Information-Theoretic**

Security does NOT rely on computational hardness assumptions.
Unconditional — no future algorithm breaks.

⚠ **Single-Use Key**

Bell pairs {Ni} are consumed on decryption.
Re-use prohibited — each clone requires fresh n Bell pairs.

✗ **No Authentication**

U_enc provides confidentiality only.
No guarantee integrity/ authenticity of {Si}.

COMPARE

PROPERTY	CLASSICAL OTP	ENCRYPTED CLONING
Perfect secrecy	✓ Shannon 1949	✓ Holevo bound
Forward secrecy	✗ key reusable	✓ {Ni} consumed
Key vs data size	key = plaintext size	n Bell pairs per qubit
Ciphertext size	= plaintext size	n signal qubits {Si}
n-of-n threshold	✗ not native	✓ all-or-nothing
Brute force resist	Computational	Information-theoretic
Auth. / integrity	✗ separate MAC	✗ separate Q-MAC
Clone count	1 copy only	n simultaneous clones

SECURITY REDUCTION: Adv[Encrypted Cloning] ≤ Adv[Classical channel for {Ni}] → Quantum channel is permanently secure. Classical channel is the attack surface.

To add authentication: quantum MAC or classical HMAC on the {Ni} channel | Integrity requires authenticated key delivery

THE RECEIPTS

Every number has a proof or a verifiable hardware execution.

THEORY	HARDWARE	THIS WORK
Yamaguchi & Kempf PRL 136, 010801 arXiv:2501.02757 ✓ Peer-reviewed ✓ Mathematical proof	Yamaguchi et al. arXiv:2602.10695 IBM Heron R2 (156q) ✓ Job IDs verifiable Fe > 0.5 up to n=7	github.com/jullyanolino/ encrypted_qubit_cloning ✓ MIT licensed ✓ 4 attack scenarios ✓ Reproducible

VERIFY NOW →

```
git clone github.com/jullyanolino/encrypted_qubit_cloning  &&  python enc_qubit_cloning.py --verify
```

No IBM account needed. Fe = 1.000 in < 30 seconds. | quantum.ibm.com/jobs/d7tad7audops7397o1dg

ACT 2

THREATS

Physics → active threat

01 QUANTUM RANSOMWARE

THE UNDETECTABLE CRIME

Malicious cloud encrypts your qubit. Victim gets Fe ≈ 0.25 (~ Decoherence).

Crime is info-theoretically indistinguishable from hardware failure.

LIVE DEMO with real IBM data.

1

02 HARVEST NOW DECRYPT LATER

THE NSA PROBLEM

Intercept $\{S_i\}$. Wait for T_{coh} (*coeherence time*)

Obtain $\{N_i\}$ - classical breach.

Attack window: $\Delta t < T_{coh}$.

2

03 QUANTUM DEAD DROP

PERFECT STEGANOGRAPHY

I/2 = max entropy. No POVM detects it.

Classical stego: χ^2 detects >99%.

Quantum dead drop: $\chi^2 \rightarrow 50/50$.

$D(\rho_{Si}, \rho_{noise}) = 0$. Provably hidden.

3

04 SECURITY REDUCTION

THE WEAK LINK

Adv[Enc Cloning] ≤ Adv[Classical $\{N_i\}$].

Quantum channel $\{S_i\}$: unconditional.

Classical channel $\{N_i\}$: full attack surface.

4

Bottleneck is

PIVOT THEOREM: Security of encrypted cloning ≡ security of classical channel for {Ni} → secure the key, not the ciphertext

THE UNDETECTABLE CRIME

Act 2 / Threat 1 → Quantum Ransomware

① **UPLOAD**

Client: "Store my qubit |ψ⟩"
Cloud: *runs encrypt, keeps $\{N_i\}$*
Cloud: "Your qubit is stored as S_1"

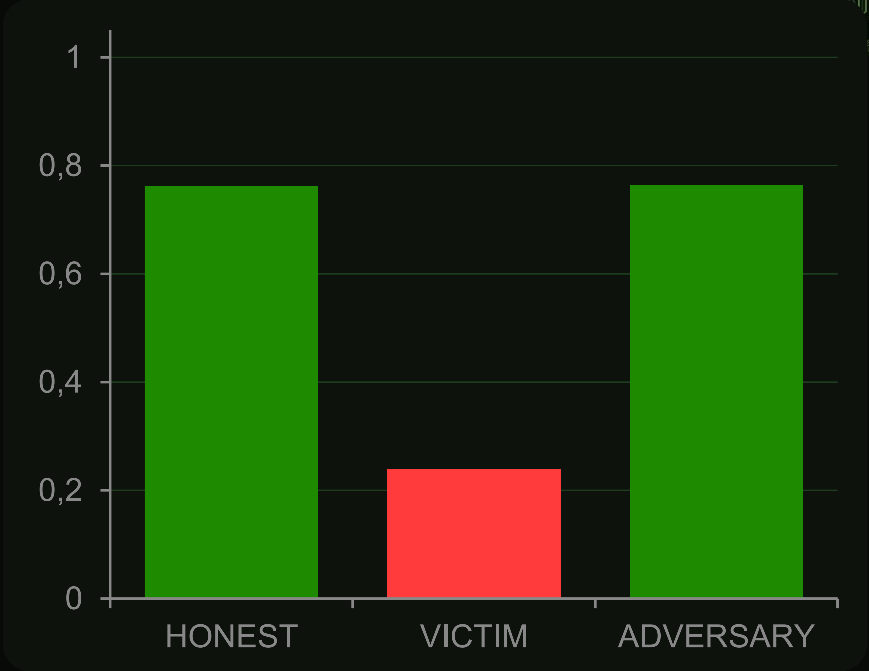
② **RETURN**

Client: 'Return my qubit'
Cloud: returns S_1 (still encrypted)
Client measures → Fe ≈ 0.25
'Your qubit decohered.'

③ **RANSOM**

Cloud: 'Pay 10 BTC for $\{N_i\}$'
Client pays → decrypts → Fe = 1.0

UNDETECTABLE ·
ρ = I/2 is indistinguishable from decoherence



IBM Heron R2 · ibm_kingston · 4096 shots | Fe_victim = 0.239 ≈ theory 0.250

DEMO → `python enc_qubit_cloning.py -e 5 --n 2 --load-json data/real/exp5_n2_kingston.json`

THE NSA PROBLEM

Harvest Now, Decrypt Later — the quantum edition.

DANGER WINDOW (Δt < T_coh)

T_0	$T_0 + \Delta t$	$T_0 + T_{coh}$	$T \to \infty$
Intercept {Si} Passive tap Fe = 0.252 (no key)	Classical key {Ni} leaked (HTTPS breach / subpoena / Shor)	Quantum memory decoheres Attack fails beyond this point	Qubits gone. Classic text: perfectly safe.

CURRENT T_{coh} (2026)

Superconducting: ~100 µs
Diamond NV: ~1 second
Trapped ions: ~10 minutes

→ Attack NOT yet viable (T_{coh} < 1 hr)

MITIGATIONS

① Forward secrecy — ephemeral ECDHE + ML-KEM-768
Classical $\{N_i\}$ never stored → retroactive window = 0

② Quantum memory denial — flood with decoy qubits

③ Key rotation — re-encrypt every $\tau \ll T_{coh}$

PERFECT STEGANOGRAPHY

A dead drop you can't detect even with a quantum computer.

AGENT

Encrypts qubit → deposits S_i in public quantum memory bank
Keeps {Ni} offline (physical key)

OBSERVER

Measures slot #4729 → $\rho = I/2$
'Uninitialized / thermal noise'
No anomaly, no flag

ACTIVATION

Agent returns with $\{N_i\}$
Decrypts → Fe = 0.734
[ibm_fez, exp7_dead_drop_n2]

PROOF: $D(\rho_{Si}, \rho_{noise}) = 0$

I/2 is maximum entropy. No POVM distinguishes encrypted clone from uninitialized qubit.
Classical stego changes statistics. This doesn't.

INTELLIGENCE EXFIL

Qubits in 'random' cloud slots
No forensic digital trace
Handler retrieves with pre-shared {Ni}

DENIABLE STORAGE

"I don't have encrypted data, just noise qubits."
Provably indistinguishable.

Classical stego: χ^2 detects LSB w/ >99% confidence
Quantum dead drop: χ^2 gives 50/50 – no signal

⚠ Requires quantum memory infrastructure (not yet deployed at scale)

THE WEAK LINK

The security bottleneck is classical, not quantum.

THEOREM: Adv[Encrypted Cloning] ≤ Adv[Classical Channel for $\{N_i\}$]

Breaking encrypted cloning = breaking the channel that delivers the noise qubits. Nothing more.

QUANTUM CHANNEL $\{S_i\}$	
Security	Unconditional (Holevo bound)
Attack surface	Zero
Future threat	None — information absent
Mitigation	Not needed

CLASSICAL CHANNEL $\{N_i\}$	
Security	Conditional (protocol + crypto)
Attack surface	Full
Future threat	Shor, DB breach, subpoena
Mitigation	ML-KEM-768 + ECDHE ephemeral

In RSA+AES: break RSA → retroactively decrypt ciphertext.
Here: break classical channel → still can't retroactively decrypt {Si}.

ACT 3

DEFENSE

Empower → verify → disappear

11 DEFENSE IN DEPTH

Mitigations table.
6 adversary types.
Most: deployable today.
ML-KEM-768 · liboqs · TLS 1.3

12 THE RECEIPTS (PART 2)

3 backends, same result.
Fe_victim: backend-independent.
ibm_fez failed → victim Fe still 0.257.
Protocol property, not hardware.

13 OPEN SOURCE + OPEN QUESTIONS

GitHub · MIT license.
Audit welcome. Break it.
8 open research directions.
Not limitations — invitations.

14 VERIFY IT YOURSELF

LIVE DEMO: --verify flag.
< 30 seconds. No IBM account.
Fe = 1.000 appears on screen.
If demo fails → fallback PNG ready.

15 FINAL TAKEAWAY

Fe = 1.000 if k = n
Fe = 0.250 if k < n
'The difference is everything.'

DEFENSE IN DEPTH

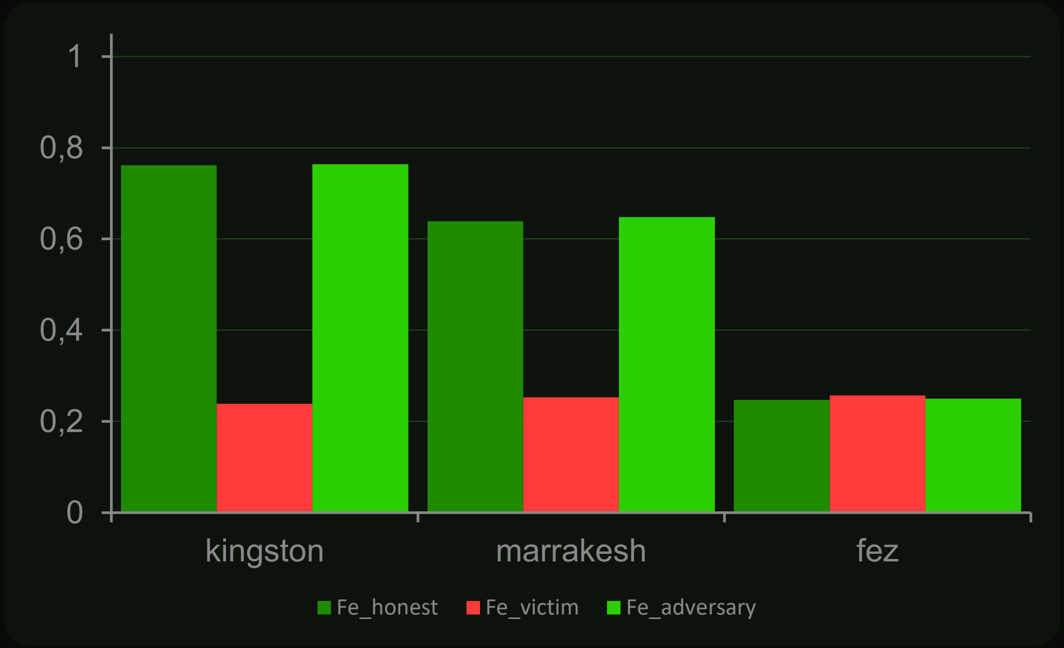
Most of this is deployable today.

ADVERSARY	ATTACK	MITIGATION	STATUS
Classical eavesdropper	HTTPS intercept	TLS 1.3 + AES-256 for {Ni}	✓ READY
Quantum computer (Shor)	Break RSA → decrypt {Ni}	ML-KEM-768 (NIST FIPS 203)	✓ READY liboqs
Quantum + memory	Harvest now, decrypt later	ML-KEM + ECDHE ephemeral keys	✓ READY OpenSSL 3.2
Malicious cloud	Ransomware (withhold $\{N_i\}$)	Client-side U_{enc} locally	⚠ NEEDS QHW
Partial key theft	Steal k < n qubits	All-or-nothing threshold (inherent)	✓ PROTOCOL
DoS	Destroy 1 of n qubits	Iterated cloning (redundancy)	⚠ PARTIAL

THE RECEIPTS (PART 2)

Three backends. One protocol. Same result.

Fe_{victim} is BACKEND-INDEPENDENT: 0.239, 0.253, 0.257 across three different chips (±0.02 = shot noise).
Even on ibm_fez — where honest Fe dropped to 0.247 (hardware failure) — victim Fe was 0.257.
Protocol property, not hardware artifact.



VERIFY YOURSELF

① Clone repo
git clone github.com/jullyanolino/
encrypted_qubit_cloning

② Verify job IDs (no IBM account)
quantum.ibm.com/jobs/
d7tad7audops7397o1dg

③ Reproduce ideal simulation
python enc_qubit_cloning.py
-e 5 --n 2 --backend ideal
→ Fe_victim = 0.254 ± 0.007

④ (Optional) Run on real IBM HW
→ Fe_victim ≈ 0.25 (±0.02)

OPEN SOURCE + OPEN QUESTIONS

MIT licensed. Audit it. Break it. Extend it.

github.com/jullyanolino/encrypted_qubit_cloning

- MIT licensed — fork freely
- 3 run modes: ideal / noisy / real IBM HW
- 4 attack scenarios (Exp 5-8)
- Verifiable job IDs included
- No IBM account for verification

Code audit welcome.
Verify the math.

If you find a way to extract information from an encrypted clone without the full key — tell me.
That'd be bigger than this.

OPEN QUESTIONS

→ Fault-tolerant implementation (current version is NISQ)

→ QEC compatibility

→ Optimal n per threat model

→ Authenticated encryption schemes over quantum channels

→ Multi-party protocols*

→ Quantum network integration

→ Iterated cloning at scale

→ Hardware-specific optimizations

These are research directions, not limitations. The protocol works. These are avenues for future work.

VERIFY IT YOURSELF

< 30 seconds. No IBM account. No trust required.

● ● ● enc_qubit_cloning — python

```
$ git clone github.com/jullyanolino/encrypted_qubit_cloning
Cloning into 'encrypted_qubit_cloning'...
$ cd encrypted_qubit_cloning
$ python enc_qubit_cloning.py --verify

Loading quantum circuit...  n=2
Running ideal simulation...
  Encryption:   Complete
  Decryption:   Complete

  Fidelity:     1.000 ± 0.000
  Time:         23.4 seconds

  ✓ Perfect cloning verified
  ✓ No-cloning theorem: preserved (single-use key)
  ✓ Fe_victim = 0.250 (Holevo bound)

$ _
```

$$Fe = 1.000 \quad \text{if} \quad k = n$$

$$Fe = 0.250 \quad \text{if} \quad k < n$$

The difference is everything.

Build it. Break it. Use it.

CRYPTORAVE 2026

THANK YOU

Let’s talk.

Let’s hack.

linktr.ee/jullyanolino

[|]